SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	Blessy S
Register Number	192421053
Class / Slot	Slot B
Capstone Title	Cloud security
Capstone Product Type	Web app / Android app / iOS app / Hybrid app
CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

CloudGuard is a cloud security posture management (CSPM) and threat-detection platform that continuously scans an organization's cloud accounts (AWS/Azure/GCP) for misconfigurations, ingests activity logs, flags anomalies with an AI model, and produces compliance reports.

Capstone Element	Student Entry
Primary users	System Administrators, Security Analysts, IT Staff, Organization Employees
User roles	Admin, Security Analyst, Employee/User, AuditorAdmin, Security Analyst, Employee/User, Auditor
Main modules	User Authentication, Access Control, Threat Detection, Data Encryption, Security Monitoring, Incident Management, Audit Logs
Cloud services identified in CO1	SaaS (Security Dashboard), PaaS (Application Development), IaaS (Virtual Machines, Storage, Networking)
API/backend need	Secure REST APIs for user authentication, threat monitoring, log management, and cloud resource access
Database need	Cloud database to store user accounts, security logs, threat reports, access permissions, and audit records
File upload/storage need	Secure cloud storage for security reports, encrypted documents, system logs, and backup files
Analytics/AI/Python module	Python-based threat detection, anomaly detection, log analysis, risk prediction, and security analytics using machine learning

Activity 2: Workload Classification

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason

Workload Type: Security Monitoring and Data Processing

Reason:
The Cloud Security application continuously monitors user activities, detects security threats, processes authentication requests, stores security logs, and analyzes network events. It requires secure cloud storage, reliable computing resources, and scalable infrastructure to ensure data protection, real-time threat detection, and high system availability.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000
Input	**Value**	**Reason / Assumption**
Registered users expected	10,000	Medium-sized organization using the cloud security platform
Active user percentage	20%	Around 20% of users are active during a normal day
Peak factor	2	Traffic approximately doubles during peak working hours
Actions per user per minute	3	Users perform login, monitoring, report viewing, and alert checking
Calculated RPS	20 RPS	(10,000 × 20% × 2 × 3) ÷ 60 = **20 Requests/Second**
Daily active users	2000	10,000 × 20% = **2,000**
Daily requests	60000	2,000 × 30 average actions per day = **60,000**

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1

Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4

Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver Selected	Score	Reason
Base web/API application	3	Handles user requests and security dashboard functions.
Authentication and roles	5	Frequent user logins, identity verification, and access control require high CPU usage.
API traffic	4	Secure APIs process requests between clients and cloud services.
Database/search/reporting	4	Stores security logs, retrieves data, and generates reports.
Background jobs	3	Performs scheduled backups, log cleanup, and security scans.
Analytics/dashboard	4	Processes dashboards, security metrics, and visualization.
AI/Python module	5	Machine learning detects threats, anomalies, and suspicious activities.
Total CPU Score	28	Sum of all CPU driver scores.
Recommended vCPU	8VCPU	Suitable for a medium-sized Cloud Security application with AI-based threat detection and moderate traffic.

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2GB
Web/API service	1-2 GB	2GB
Small database	2-4 GB	4GB
Medium database/cache	4-8 GB	6GB
Analytics/reporting	4-8 GB extra	4GB
AI/Python module	4-16 GB extra	8GB
25% buffer	Subtotal x 0.25	7GB
Final RAM	Round up to standard size	32GB

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	30 GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	50 GB
File uploads	Avg file MB x files/day x days x 1.3	100 GB
Logs	Requests/day x log KB x days / 1024 MB	30 GB
Backup/growth buffer	Subtotal x 30%	63 GB
Final storage	Round up to standard disk size	256 GB SSD

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	No
Two-tier design	App VM + managed/separate database	Considered, but not selected
Three-tier design	Frontend, backend/API and database separated	selected
Containerized backend	Microservices, portable API, CI/CD	Future enhancement

Serverless function	Event-based task, notification, small automation	Used for alerts and notifications
Managed database/storage	Reliability, backup and scaling required	Selected

Final VM Plan: selected plan, vCPU, RAM, storage and reason

Adopted plan: a containerized three-tier architecture backed by managed services. Frontend: 2 vCPU / 4 GB (static hosting/CDN is acceptable at scale). Backend/API tier: 4 vCPU / 8 GB per instance, 2 instances behind a load balancer to cover the ~10 RPS peak with headroom. AI/analytics tier: a dedicated 8 vCPU /32 GB container service (GPU optional later) to isolate the heaviest CPU score driver. Database: managed relational database service, sized independently rather than as a VM. Storage: 80 GB SSD system disk per compute node plus a 500 GB+ auto-scaling object storage bucket for reports and log archives. Reason: this matches the calculated CPU score (11 → scale out) and RAM total (~35 GB) while keeping the database reliable and the AI workload isolated from user-facing traffic.

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment

Context	Selected Approach	Justification
Local prototype/testing	Type 2 Hypervisor	Used for development and testing on a local computer using VirtualBox or VMware Workstation.
Cloud pilot deployment	Cloud-managed virtualization	Cloud provider manages the hypervisor, making deployment simple, secure, and scalable.
Production growth scenario	Type 1 Hypervisor	Runs directly on physical hardware, providing high performance, security, and reliability for production workloads.
Module suitable for container	Container Runtime	API service, authentication, threat detection, and monitoring modules can run efficiently in containers for portability and scalability.

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Before every production deployment	Keep for **30 days** for rollback if deployment fails
Before database schema change	Snapshot	Before every database update	Keep for **14 days** to recover database changes
Before OS/package update	Snapshot	Before every database update	Keep for **14 days** in case the update causes issues
Before review/demo	Snapshot	One day before presentation/demo	Keep until the demo is completed
Create testing environment	Clone	Whenever a new testing environment is required	Delete after testing to save storage
Create team demo copy	Clone	Before team demonstrations or training	Keep until the demonstration is completed

Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	**VM / Container / Managed Service**	**Reason**
Frontend	Container	Easy deployment, scalability, and fast updates for the web interface.
Backend/API	Container	Secure REST APIs can scale independently and are easy to manage..
Database	Managed Service	Provides automatic backup, high availability, security, and scalability.
File storage	Managed Service	Secure cloud object storage for logs, reports, and encrypted files.
AI/Python module	Container	Runs threat detection and machine learning models independently with easy scaling.
Analytics/dashboard	Container	Supports real-time security monitoring and dashboard updates efficiently.
Notification/background jobs	Serverless Function	Executes scheduled security scans, sends alert emails/SMS, performs automated backups, log rotation, and incident notifications without keeping a server running.

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

Cloud Guard is classified as a hybrid database-heavy, analytics and AI workload, not a simple CRUD app, because it continuously ingests cloud logs, scores risk, and serves compliance dashboards. Traffic calibration from 500 registered users yields roughly 200 concurrent users and 10 RPS at peak, driving a CPU score of 11, which the workbook guidance maps to scaling out rather than one large VM. There commended plan is a three-tier, containerized architecture: a 2 vCPU/4 GB frontend, a 4 vCPU/8 GBAPI tier running two load-balanced instances, and a dedicated 8 vCPU/32 GB AI/analytics service that isolates the heaviest compute driver from user traffic. RAM calibration totals about 35 GB across these after the 25% buffer, provisioned as two 32 GB nodes rather than one oversized machine. Storage calibration totals roughly 64 GB for OS, metadata and logs, rounded up to an 80 GB system disk, with reports, exports and raw log archives pushed to an auto-scaling object storage bucket starting at 500GB. Virtualization moves from a Type 2 hypervisor for local development, to cloud-managed VMs for the pilot, to cloud-managed VMs running a container runtime for production, since the API, scan workers and AI service are all stateless and portable. Snapshots protect rollback points before deployments ,schema changes, patching and demos, while clones are reserved for QA testing and stakeholder demo sand are never relied on as the sole backup. The database and object storage are deliberately kept as managed services rather than self-hosted VMs to guarantee backup and durability for sensitive security findings. Overall, this plan starts small enough for a capstone pilot but leaves a clear, justified growth path to auto scaling and managed services as client accounts increase.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.